# 12AL130
# 12ALZ130

AMARON® QUANTA
LIFE UNINTERRUPTED

## AMARON QUANTA™

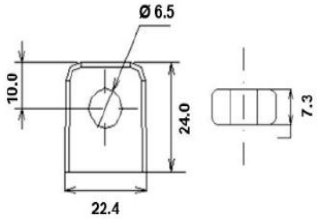
The industrial segment SMF-VRLA (Valve Regulated Lead Acid) battery for UPS applications is built to perform.

In short, the lifeline to your UPS applications

*AMARON QUANTA* ™ is a product of fail- safe, fool-proof battery technology, produced and tested in our premier manufacturing facility. Built to the highest technical competence in its class, the QUANTA is an example of Amara Raja's commitment to bringing the best of technology to your table

It features several firsts for the battery industry like the unique Radgrid™

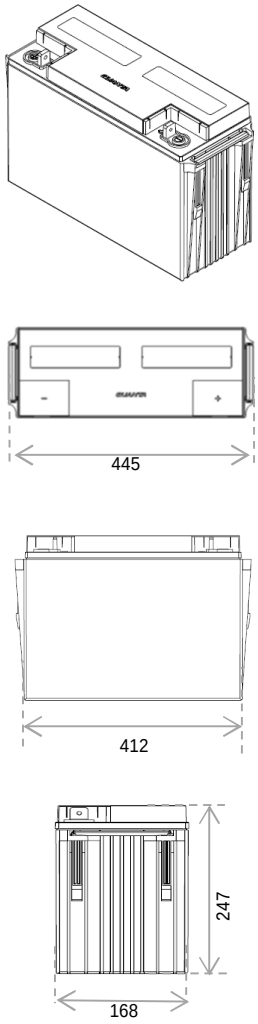
## TERMINAL LAYOUT




**All dimensions are in mm**

## Specification

| | | |
|---|---|---|
| Nominal Voltage | | 12V |
| Rated Capacity | | 130Ah / $C_{20}$Hr / 1.75VPC / 27°C |
| Dimensions (±2mm) | Length | 445 mm (17.52 in.) |
| | Width | 168 mm (6.61 in.) |
| | Total Height | 247 mm (9.72 in.) |
| Weight (±5%) | | 42.0 Kg (92.59 lbs) |
| Terminal Type | | M6 x 25mm Bolted |
| Capacity @ 27°C | 121.00 Ah | (10hr, 12.10 A, 10.5 V/battery) |
| | 111.50 Ah | (5hr, 22.30 A, 10.2 V/battery) |
| | 84.42 Ah | (1hr, 84.42 A, 9.6V /battery) |
| Capacity affected (Temperature at C20 hr rate) | 40°C (104°F) | 110% |
| | 27°C (80.6°F) | 100% |
| | 0°C (32°F) | 80% |
| | -15°C (5°F) | 60% |
| Case Material | Standard | PPCP (12AL130) |
| | FR Version | UL 94-V0 (12ALZ130) |
| Internal Resistance (IR) | | Approx. 4.54 mΩ for a fully charged battery (27°C) |
| Short Circuit Current (As per IEC) | | 2803 A |
| Operating Temp. range | | -20°C to +60°C (50 to 60°C for shorter duration) |
| Nominal Operating Temp. range | | 27°C ± 3°C |
| Standby use (27°C) | Charging Voltage | 13.5 V/battery |
| | Charging Current | Max. 25% of rated capacity |
| | Temp. Compensation | ± 18mV/battery/°C |
| Cyclic use (27°C) | Charging Voltage | 13.8 V/battery |
| | Charging Current | Max. 25% of rated capacity |
| | Temp. Compensation | ± 30mV/battery/°C |
| Self-Discharge | | < 4% per month at 27°C |




### Constant Power Discharge Rating (Watts Per Battery) @ 27°C *

| ECV/ Time | 10min | 15min | 20min | 30min | 60min | 2hrs | 3hrs | 5hrs | 8hrs | 10hrs | 20hrs |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.60 | 3128 | 2602 | 1965 | 1608 | 919 | 557 | 403.0 | 279.0 | 183.0 | 148.0 | 79.0 |
| 1.65 | 3073 | 2543 | 1945 | 1558 | 904 | 549 | 400.0 | 274.5 | 179.0 | 147.5 | 78.5 |
| 1.70 | 3017 | 2483 | 1924 | 1508 | 889 | 541 | 397.0 | 270.0 | 176.5 | 147.0 | 78.0 |
| 1.75 | 2886 | 2365 | 1888 | 1482 | 861 | 532 | 394.0 | 266.0 | 173.0 | 146.0 | 77.5 |
| 1.80 | 2755 | 2247 | 1852 | 1456 | 833 | 522 | 391.0 | 262.0 | 170.0 | 145.0 | 77.0 |

### Constant Current Discharge Rating (Amperes) @ 27°C *

| ECV/ Time | 10min | 15min | 20 min | 30min | 60 min | 2 hrs | 3 hrs | 5 hrs | 8hrs | 10 hrs | 20 hrs |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.60 | 293.59 | 234.66 | 186.25 | 141.30 | 84.42 | 51.20 | 34.99 | 24.09 | 14.94 | 12.64 | 6.96 |
| 1.65 | 284.70 | 224.60 | 179.40 | 135.70 | 83.40 | 49.10 | 33.90 | 23.20 | 14.60 | 12.50 | 6.80 |
| 1.70 | 275.72 | 214.52 | 172.64 | 130.00 | 82.28 | 47.05 | 32.88 | 22.30 | 14.29 | 12.33 | 6.61 |
| 1.75 | 265.31 | 209.68 | 171.05 | 128.71 | 81.25 | 46.43 | 32.83 | 22.22 | 14.21 | 12.10 | 6.50 |
| 1.80 | 260.00 | 206.35 | 166.67 | 127.45 | 77.75 | 46.10 | 32.66 | 22.03 | 14.13 | 12.04 | 6.47 |

Note:
1. The above data are average values per battery and can be obtained within five charge/discharge cycle
2. A tolerance of ±5% is applicable for the above constant power discharge and constant current discharge values.
3. Recommended to follow IEEE -485 Standard for Battery sizing (In terms of Aging Margin, Design Margin) for Optimal Performance & Life.
4. Considerable Voltage drop across cables, if any shall be considering during battery sizing.

AMARA RAJA
Gotta be a better way

**Design improvement is a continuous process of Amara Raja. As a result, specifications are subject to change without prior notice

QUANTA™
– WHEN ALL ELSE FAILS

**CAUTION**

- Avoid short circuit
- Don't charge in a sealed container





**THE LONG LIFE UPS BATTERY**

## Performance

A clutch of design features ensures that AMARON QUANTA™ batteries perform predictably and reliably every time

- ✓ Proven AGM technology that ensures maintenance free characteristics
- ✓ A unique heavy duty corrosion-resistant alloy for positive grids to increase cyclic life in tropical
- ✓ Radgrid™ profile provides lower internal resistance and superior high-discharge performance
- ✓ Instacharge™ a patented paste recipe for excellent charge acceptance
- ✓ Low self-discharge rates for extended storage periods
- ✓ Design Float life of upto 10 years
- ✓ Clean and Sleek looks

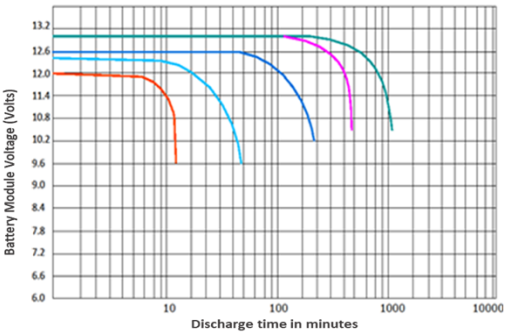
## Compliance

- ✓ JIS C 8702 Certified
- ✓ UL (UL-1989) & CE Certified
- ✓ Complies to IEC61056 & EUROBAT
- ✓ Complies to IS 16220
- ✓ Manufactured in ISO 9001, ISO 14001, ISO45001:2018 certified facilities

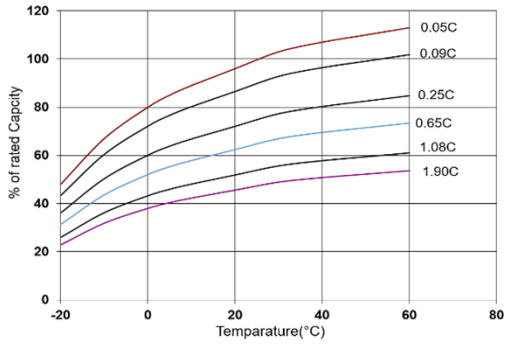
## Applications

- ✓ Data Centers
- ✓ Banks & Financial Markets
- ✓ Network Operations Centers
- ✓ Industrial Process Control Facilities
- ✓ Safety, Surveillance & Security Systems
- ✓ Semiconductor Manufacturing
- ✓ Power Generation Plants
- ✓ Hospital & Testing laboratories

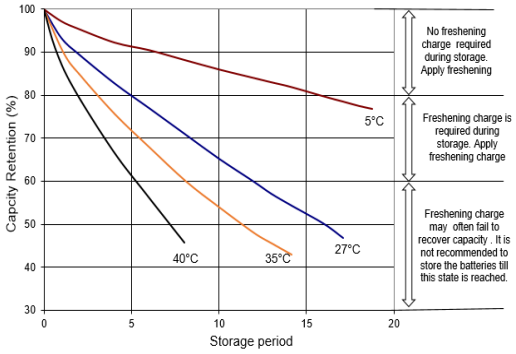
### Discharge Characteristics



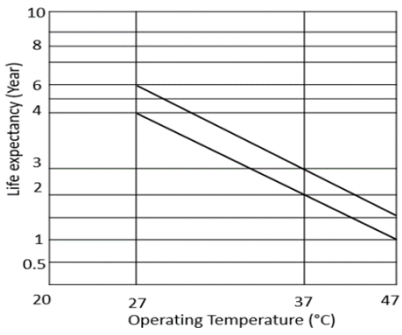

### Temperature Effect on Capacity




### Shelf Life Characteristics




### Float Service Life vs. Temperature





**AMARA RAJA ENERGY & MOBILITY LIMITED**
(Formerly Known as Amara Raja Batteries Limited)

- **Corporate Operations Office:**
  Terminal A, 1-18/1/AMR/NR, Nanakramguda, Gachibowli, Hyderabad-500032, India, E-Mail: mktg@amararaja.com
  www.amararajaeandm.com
- **Registered Office & Manufacturing Facility-1:**
  UNIT-I,Karakambadi - 517520, Tirupati, Andhra Pradesh, India, TEL: +91-877-2265000, FAX: +91-877-2285600
- **Manufacturing Facility-2**
  UNIT-II, Nunegundlapalle, Bangarupalyam,Chittoor - 517416. Andhra Pradesh, India


AMARA RAJA
Gotta be a better way

*ARE&M/IAE/AQ/130Ah; Oct'2023, Rev-01*

**Design improvement is a continuous process of Amara Raja. As a result, specifications are subject to change without prior notice